

● EASY

● ALGEBRA

● ANSWER KEY

Algebra

10 answers · Rationale-rich · Teach from doc

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Q1**A**

A. $1.87c + 2.40t \leq 15$

Choice A is correct. It's given that each container costs \$ 1.87 and c represents the number of containers. It follows that $1.87c$ represents the cost, in dollars, of c containers. It's also given that each roll of tape costs \$ 2.40 and t represents the number of rolls of tape. It follows that $2.40t$ represents the cost, in dollars, of t rolls of tape. Therefore, the total cost, in dollars, for c containers and t rolls of tape is $1.87c + 2.40t$. It's given that Leo has \$ 15. It follows that the total cost, in dollars, must be less than or equal to 15. Therefore, the inequality $1.87c + 2.40t \leq 15$ represents this relationship.

Choice B is incorrect. This inequality represents a relationship where Leo has at least \$ 15, not at most \$ 15.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**D**

D. $12.50g + 1.50t = 80$

Choice D is correct. Since a 10-ride pass costs \$12.50 and g is the number of 10-ride passes Tony buys in a month, the expression $12.50g$ represents the amount Tony spends on 10-ride passes in a month. Since a single-ride pass costs \$1.50 and t is the number of single-ride passes Tony buys in a month, the expression $1.50t$ represents the amount Tony spends on single-ride passes in a month. Therefore, the sum $12.50g + 1.50t$ represents the amount he spends on the two types of passes in a month. Since Tony spends a total of \$80 on passes in a month, this expression can be set equal to 80, producing $12.50g + 1.50t = 80$.

Choices A and B are incorrect. The expression $g + t$ represents the total number of the two types of passes Tony buys in a month, not the amount Tony spends, which is equal to 80, nor the cost of one of each pass, which is equal to $1.50 + 12.50$. Choice C is incorrect and may result from reversing the cost for each type of pass Tony buys in a month.

Q3**A**

A. 11

Choice A is correct. Dividing both sides of the given equation by 8 yields $x = 11$. Therefore, 11 is the solution to the given equation.

Choice B is incorrect. This is the solution to the equation $x + 8 = 88$.

Choice C is incorrect. This is the solution to the equation $x - 8 = 88$.

Choice D is incorrect. This is the solution to the equation $\frac{x}{8} = 88$.

Q4**D**

D. $mx = 5.7x$

Choice D is correct. It's given that Juan rides his bike at an average rate of 5.7 minutes per mile. The number of minutes it will take Juan to ride x miles can be determined by multiplying his average rate by the number of miles, x , which yields $5.7x$. Therefore, the function $mx = 5.7x$ models the number of minutes it will take Juan to ride x miles.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q5**23**

The correct answer is 23. Since it's given that $x = 2$, the value of y can be found by substituting 2 for x in the first equation and solving for y . Substituting 2 for x yields $3(2) + y = 29$, or $6 + y = 29$. Subtracting 6 from both sides of this equation yields $y = 23$.

Q6**B**

B. If the rectangle has length 8 and width 3, its perimeter is less than or equal to 27.

Choice B is correct. It's given that a rectangle has length l and width w , and the inequality $2l + 2w \leq 27$ gives the possible values of l and w for which the perimeter of this rectangle is less than or equal to 27. To determine the best interpretation of $(l, w) = (8, 3)$ in this context, the values can be substituted in the given inequality. Substituting 8 for l and 3 for w in this inequality yields $2(8) + 2(3) \leq 27$, which is equivalent to $16 + 6 \leq 27$, or $22 \leq 27$. Since this inequality is true, if the rectangle has length 8 and width 3, its perimeter is less than or equal to 27.

Choice A is incorrect. The interpretation of $(l, w) = (8, 3)$ implies that the rectangle has length 8 and width 3, not length 3 and width 8.

Choice C is incorrect. The interpretation of $(l, w) = (8, 3)$ implies that the rectangle has length 8 and width 3, not length 3 and width 8.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7**D**

D. $x + 3y = 70$

Choice D is correct. Since x represents the number of 1-minute segments and y represents the number of 3-minute segments, the total length of the video is $1 \cdot x + 3 \cdot y$, or $x + 3y$, minutes. Since the video is 70 minutes long, the equation $x + 3y = 70$ represents this situation.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q8**24**

The correct answer is 24. Multiplying both sides of the given equation by 3 yields $3(3x + 2) = 24$. Using the distributive property to rewrite the left-hand side of this equation yields $9x + 6 = 24$.

Q9**A**

A. With each monthly deposit, the amount in Hana's bank account increased by \$ 25.

Choice A is correct. It's given that t represents the number of monthly deposits. In the given function $ft = 100 + 25t$, the coefficient of t is 25. This means that for every increase in the value of t by 1, the value of ft increases by 25. It follows that with each monthly deposit, the amount in Hana's bank account increased by \$ 25.

Choice B is incorrect. Before Hana made any monthly deposits, the amount in her bank account was \$ 100.

Choice C is incorrect. After 1 monthly deposit, the amount in Hana's bank account was \$ 125.

Choice D is incorrect and may result from conceptual errors.

Q10**A**

A. 1

Choice A is correct. The first equation in the given system of equations is $x = 4$. Substituting 4 for x in the second equation in the given system of equations yields $y = 5 - 4$, or $y = 1$.

Choice B is incorrect. This is the value of x in the solution to the given system of equations, not the value of y .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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